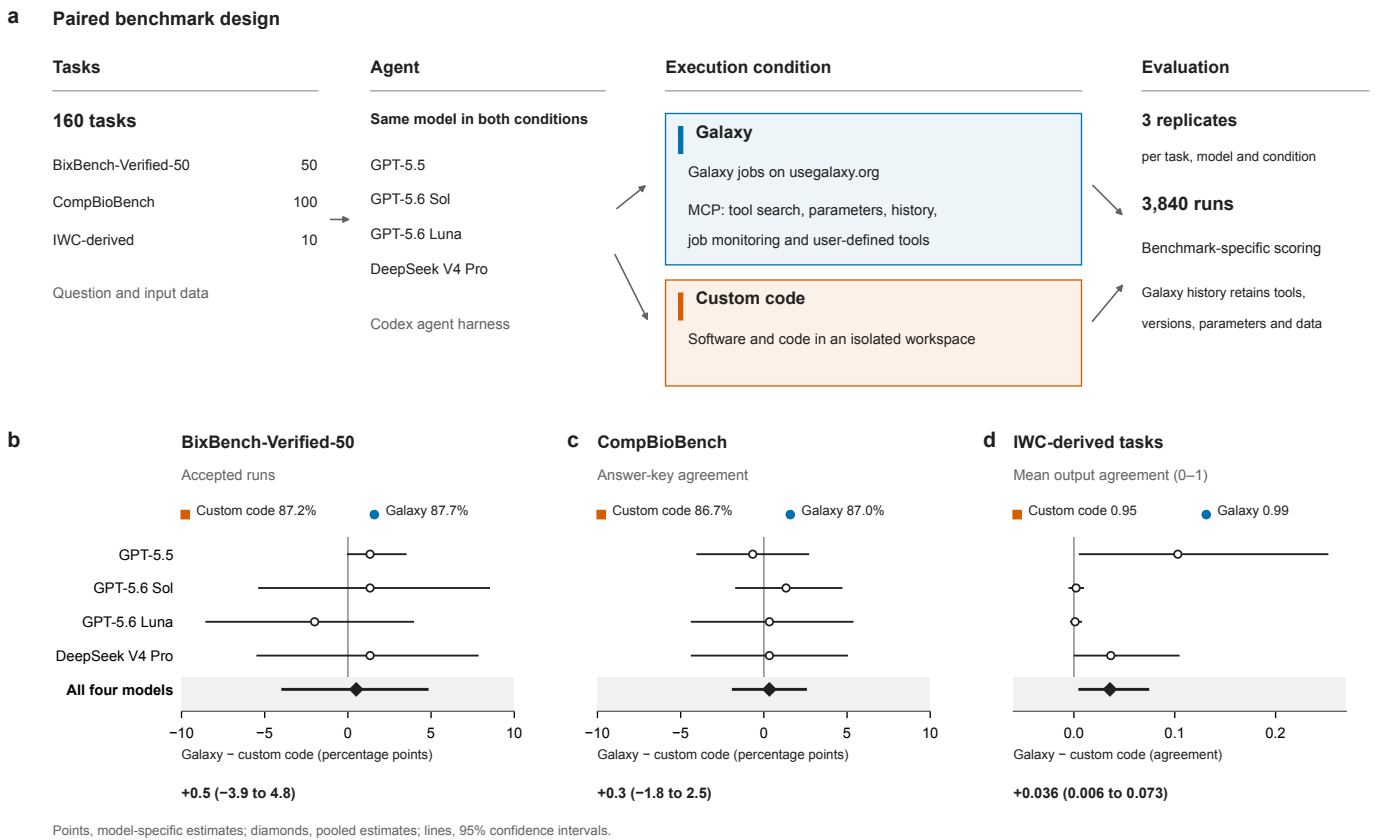

a
Paired benchmark design
Tasks
160 tasks
BixBench-Verified-50 50
CompBioBench 100
IWC-derived 10
Question and input data
Agent
Same model in both conditions
GPT-5.5
GPT-5.6 Sol
GPT-5.6 Luna
DeepSeek V4 Pro
Codex agent harness
Execution condition
Galaxy
Galaxy jobs on usegalaxy.org
MCP: tool search, parameters, history, job monitoring and user-defined tools
Custom code
Software and code in an isolated workspace
Evaluation
3 replicates
per task, model and condition
3,840 runs
Benchmark-specific scoring
Galaxy history retains tools, versions, parameters and data
b
BixBench-Verified-50
Accepted runs
Custom code 87.2%
Galaxy 87.7%
GPT-5.5
GPT-5.6 Sol
GPT-5.6 Luna
DeepSeek V4 Pro
All four models
−10 −5 0 5 10
Galaxy − custom code (percentage points)
+0.5 (−3.9 to 4.8)
c
CompBioBench
Answer-key agreement
Custom code 86.7%
Galaxy 87.0%
−10 −5 0 5 10
Galaxy − custom code (percentage points)
+0.3 (−1.8 to 2.5)
d
IWC-derived tasks
Mean output agreement (0–1)
Custom code 0.95
Galaxy 0.99
0.0 0.1 0.2
Galaxy − custom code (agreement)
+0.036 (0.006 to 0.073)
Points, model-specific estimates; diamonds, pooled estimates; lines, 95% confidence intervals.